OMAR EL TERRAS

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EDUCATION

Bachelor of Engineering Science B.E.Sc. Software Engineering (3.8 GPA) *Expected Graduation: April 2026*
The University of Western Ontario, London, ON

Coursework: Algorithms & Data Structures, Database Systems, Object-Oriented Programming, Machine Learning

EMPLOYMENT EXPERIENCE

Medical Imaging Software Developer *October 2025 - Present*
Robarts Research Institute, London, ON

- Developed a Python-based medical imaging application implementing a patented coronary blood-flow algorithm to compute FFR from 4DCT studies, reducing manual analysis time by 40% while improving clinical review efficiency.
- Engineered backend modules for curve fitting, ROI extraction, DICOM loading, and FFR computation, boosting performance by 30% across Siemens, Canon, and GE datasets with scanner-aware and vectorized processing.
- Designed and built a multi-view dark-mode UI with CTA visualization, time-enhancement curves, fitted models, and instantaneous FFR plots, creating a workflow that streamlined researcher interaction and reduced testing overhead.
- Created reproducible output and validation pipelines, JSON result storage, automated reports, activity-tracking logs, and optimized cardiovascular flow models, reducing algorithmic discrepancies by 25% during verification.

Software Developer (QA Automation) Intern *May 2024 - August 2025*
J.D. Power, London, ON

- Developed and maintained 2,500+ automated test cases across 50+ microservices using Serenity BDD, Cucumber, JUnit, REST APIs, validating services built on Spring Boot architectures, increasing automation coverage by 30% and reducing manual regression by 20+ hours per release.
- Built automation frameworks from scratch and validated REST APIs (GET/POST/PUT) and SQL data integrity using Postman, debugging tools, Git-based version control, and CLI tools, improving defect detection by 15% while contributing fixes to backend services when issues were identified.
- Integrated automated tests into CI/CD pipelines (Jenkins, GitLab CI, Maven, Gradle), monitored pipeline health, debugged build and test failures, and ensured 100% successful deployments to staging and production environments.
- Used AI tools for development productivity, including GitHub Copilot, ChatGPT, Claude, and LLM-based tools for test generation, refactoring, code reviews, debugging assistance, API analysis, and accelerating test design.
- Collaborated on QA workflow improvements by automating 400+ Jira/Xray test cases and maintaining Requirements Traceability Matrices, while working in Agile sprints, participating in peer code reviews, and mentoring a new intern, reducing onboarding time by 50%.

PROJECTS

LevelUp | React, Node.js, PostgreSQL, Large Language Models (LLMs) *September 2025 - Present*

- Developed a cloud-ready web app that ingests unstructured data (job descriptions, resumes) and uses LLMs to generate resumes, cover letters, and mock interviews, projected to reduce drafting time by 80%.
- Designed the system architecture and core features, including 90%+ accurate PDF/DOCX resume parsing, AI-based suggestions, autosave, and job description ingestion to deliver accurate, job-aware recommendations.
- Led a 4-member Agile team, driving backlog priority, acceptance criteria, and sprint planning to address inconsistent delivery, completing a predictable and successful Sprint 1 (20 story points).
- Built a rubric-based mock interview module with secure recording, Firebase Auth, and RBAC after recognizing weak feedback tools and privacy gaps, improving user interview performance by 30%+ while strengthening data protection.

Cheer OLLI Website | HTML, CSS, JavaScript, React, MongoDB, Google Cloud *March 2024*

- Selected as the top site out of 50 teams by collaborating with nonprofit organization Cheer and leading a 4-member team to design and launch their official website.
- Increased user satisfaction from 70% to 93% by developing a full-stack React, Node.js, and MongoDB (NoSQL) platform with admin management, scheduling, and community tools through data-driven Scrum design.
- Improved scalability and team productivity by containerizing the app with Docker for deployment on GCP and streamlining collaboration with Jira and Confluence, boosting output by 25%.

TECHNICAL SKILLS

- ***Languages & Databases:*** Java, SQL, Python, JavaScript, MySQL, MSSQL, Cassandra
- ***Backend, APIs & Testing:*** REST APIs, Microservices, Serenity BDD, Cucumber, JUnit, Playwright, Selenium
- ***DevOps, Build & Cloud:*** Git, GitLab, Docker, AWS, GCP, Kubernetes, Maven, Gradle, Jenkins, CI/CD pipelines,
- ***Tools & Collaboration:*** Postman, Jira, Xray, Confluence, Grafana